Yihao Xu

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Research Profile

- Computational design science researcher working on multimodal learning and predictive analytics for digital platforms, with additional work in computational measurement of organizational culture.
- First author of a paper at the *International Conference on Information Systems* (2025) and an article in *IEEE Sensors Journal*; the ICIS paper was a Best Complete Paper Nominee / Best-in-Track Paper.
- Experience designing multimodal and multi-view learning models, modeling temporal distribution shift, and assessing organizational culture from text; Python, PyTorch, and SQL.
- Teaching assistant across two fall terms; coordinated six study sessions involving 60 participants who are deaf or hard of hearing in an AR accessibility study.

Education

M.S. in Management Science and Engineering **Expected 2027**
School of Management, University of Science and Technology of China (USTC) *Hefei, China*
Advisor: Prof. Hefu Liu.

B.M. in Information Management and Information Systems **2024**
School of Management, Hefei University of Technology (HFUT) *Hefei, China*
Selected coursework (HFUT grades):
Linear Algebra (96/100, A), Operational Research (95/100, A), Business Data Analytics (91/100, A).

Research Interests

Topics: Digital platforms and crowdfunding · Cultural and creative markets · Organizational culture · Accessible human–AI interaction
Methods: Multimodal representation learning · Multi-view deep clustering · Temporal generalization · Computational text measurement

Published Research

1. Subtly There, Deeply Felt: Modeling Cultural Values in Crowdfunding Success Prediction
Yihao Xu, Wenxiang Zhao, and Xiaoyu Yao.
46th International Conference on Information Systems (ICIS), Sharing Economy, Platforms, and Crowds Track, Nashville, TN, 2025. *ICIS 2025 Best Complete Paper Nominee / Best-in-Track Paper.*

- Creative projects convey cultural meaning through text, images, and metadata. This study models those values and examines their roles in crowdfunding success.
- The Cultural Value-based Prediction Model (CV-PM) organizes content features around Throsby's six cultural values. Value-specific convolutional layers refine noisy features, and attention learns their joint predictive contributions.
- On 14,539 Kickstarter projects, AUC reached 94.72% in Art and 95.14% in Design, exceeding the strongest respective baselines by 1.49 and 3.65 percentage points.
- Value ablations showed the largest accuracy losses for social value in Art (7.17 percentage points) and aesthetic value in Design (5.19), revealing category differences in their predictive roles.

2. Multisensor Deep Clustering Networks With Joint Learning Strategy for Fault Diagnosis Considering Consensus-Specific Constraint
Yihao Xu, Mingfeng Lu, Gang Wang, and Fang Fang.
IEEE Sensors Journal, 25(11), 19921–19936, 2025. doi:10.1109/JSEN.2025.3560460.

- Unsupervised fault diagnosis faces two problems: separately learned features may not suit clustering, and multiple sensors contain shared as well as sensor-specific signals.
- MsDCN combines a sensor-aware attentive encoder for shared patterns with sensor-specific encoders. A consensus-specific constraint balances these representations during joint optimization of reconstruction and clustering.
- Across 20 runs per dataset, MsDCN achieved the highest mean clustering accuracy on six gearbox and bearing datasets. Gains over the strongest comparison method were 5.74–11.35 percentage points on the two DIRG bearing datasets; ablations examined the sensor-attention and consensus-specific designs.

Conference Presentation

Yihao Xu, Wenxiang Zhao, and Xiaoyu Yao. *Subtly There, Deeply Felt: Modeling Cultural Values in Crowdfunding Success Prediction.* 46th International Conference on Information Systems (ICIS), Sharing Economy, Platforms, and Crowds Track, Nashville, TN, Dec. 2025.

Work in Progress

3. Multimodal Learning under Evolving Environments for Crowdfunding Outcome Prediction

Yihao Xu, Wenxiang Zhao, and Hefu Liu. *Planned submission to Information Systems Research in 2026.*

- Campaign text and images convey both overlapping and complementary information. SSC-TAP organizes shared, modality-specific, and interaction information within a common semantic reference, then learns their joint contribution to pre-launch prediction.
- Historical content–outcome relationships can change as crowdfunding environments evolve. Temporal relevance adaptation separates changes in content distributions from changes in predictive relevance, adjusting content contributions for each new campaign and its actual launch time.

4. From Anchor to Evidence: A Theory-Driven Model for Newly Launched Crowdfunding Success Prediction

Wenxiang Zhao, **Yihao Xu**, and Xiaoyu Yao. *Working Paper.*

- Models the distinct roles of project summaries and detailed information in pre-launch crowdfunding prediction, drawing on anchoring-and-adjustment theory.

5. When Differences Drive Synergy: Cultural Complementarity as a Resolution to the Efficiency–Resilience Paradox in Supply Chains

Along Liu, Mengru Zhang, **Yihao Xu**, and Hefu Liu. *Research in progress.*

- Investigates cultural complementarity in the efficiency–resilience trade-off. A scale-anchored text measurement protocol combines semantic retrieval, expert-reviewed annual-report examples, and item-level scoring.
- Links firm-level cultural profiles to disclosed supplier and customer relationships, constructing transaction-weighted matching measures for annual panel and disruption–recovery analyses.

6. Knowing When to Be Firm: Communication Style Shifts in AI-Assisted Food Delivery

Early-stage Research Project.

- Food-delivery customers may have explicit coupon requests alongside unspoken emotional needs. This project examines shifts between assertive and nonassertive AI language during coupon assistance and emotional support, tracing customer responses across conversational turns and preference profiles to study subsequent purchase decisions.

Research and Teaching Experience

Research Assistant, University of Science and Technology of China *Jun. 2026–Present*

Supported an AR smart-glasses study on communication access and social inclusion for people who are deaf or hard of hearing; coordinated materials, participant arrangements, device setup, and six study sessions involving 60 participants.

Teaching Assistant, University of Science and Technology of China *Fall 2025, 2026*

Information Technology and Organizational Strategy 42 students; 40 hours

Instructor: Prof. Juan Qin; coordinated assignments, group activities, and feedback.

Awards and Scholarships

Awards

- Outstanding Graduate of Anhui Province (top 3% students) 2024
- Outstanding Student Award (top 10% students) 2023

Scholarships

- China National Scholarship 2025
- First-Class Graduate Scholarship, USTC 2025, 2026
- Shishengqing Scholarship, School of Management, HFUT 2024
- Academic Progress Scholarship, School of Management, HFUT 2022
- First-Class Scholarship, HFUT (top 3% students) 2021

Internships

Research Intern *Jun.–Aug. 2026*

USTC Yangtze River Delta Science and Technology Strategic Frontier Research Center

Conducted desk research, case analysis, and comparative studies of firms' early development and commercialization in commercial spaceflight, AI, and semiconductors.

Technical Service Intern, CEO Office, Volkswagen (Anhui) Automotive Co., Ltd. *Apr.–Jul. 2024*

Hefei, China. Built and maintained an automotive after-sales database for recurring service reporting and operational analysis.

Computational Skills and Languages

Computational Methods: Multimodal learning · Multi-view deep clustering · Attention mechanisms · Temporal generalization · Sentence embeddings and semantic retrieval

Programming: Python · PyTorch · SQL · Stata

Languages: Chinese (native) · English (IELTS Academic 7.5)